

Assignment#1
Probability Models
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Problem 1: Naive Bayes for Model A - Support-related features

Customer Churn Using
Observed Data

Background: You are a data scientist working for a subscription-based company. Your goal is to predict whether a customer will churn ($Y=1$) or not churn ($Y=0$) using behavioral data.

($X_1 = 1$) Customer opened many support tickets last month
($X_2 = 1$) Customer spent a lot of time on help/FAQ page

Training Dataset (Historical sample)

Churn (Y)	$X_1 = 1$	$X_1 = 0$	Total
Y = 1	24	6	30
Y = 0	14	56	70

Churn (Y)
Y = 1
Y = 0

Laplace Smoothed alpha = 1

Churn (Y)	$X_1 = 1$	$X_1 = 0$	Total
Y = 1	0.78125	0.21875	1
Y = 0	0.20833333	0.79166667	1

Laplace S

Churn (Y)
Y = 1
Y = 0

Historical Churn:

$$P(Y=1) = 0.3$$

$$P(Y=0) = 0.7$$

Question A1 - Estimate Probabilities

$$P(X_1=1 \mid Y=1) = 0.78125$$

$$P(X_1=1 \mid Y=0) = 0.20833333$$

$$P(X_2=1 \mid Y=1) =$$

$$P(X_2=1 \mid Y=0) =$$

Question A2 - Normalized Posterior Probabilities

$$P(Y=1 \mid X_1=1, X_2=1) = 0.19042969$$

$$\text{Normalized: } 0.8703918$$

$$P(Y=0 \mid X_1=1, X_2=1) =$$

$$\text{Normalize}$$

Question A3 - Prediction

1. Does the model predict churn or no churn?

2. What is the predicted probability?

Model B - Pricing and competition features

($Z_1 = 1$) Customer changed pricing plan last month

($Z_2 = 1$) Customer received a competitor's marketing email

Training Dataset (Historical Sample)

Churn (Y)	$Z_1 = 1$	$Z_1 = 0$	Total
Y = 1	18	12	30
Y = 0	21	49	70

Churn (Y)
Y = 1
Y = 0

Laplace Smoothed alpha = 1

Churn (Y)	$Z_1 = 1$	$Z_1 = 0$	Total
Y = 1	0.59375	0.40625	1
Y = 0	0.30555556	0.69444444	1

Laplace S

Churn (Y)
Y = 1
Y = 0

Question B1 - Estimate probabilities

$P(Z_1=1 \mid Y=1) =$	0.59375	$P(Z_2=1 \mid Y=1) =$
$P(Z_1=1 \mid Y=0) =$	0.30555556	$P(Z_2=1 \mid Y=0) =$

Question B2 - Posterior probabilities

$P(Y=1 \mid Z_1=1, Z_2=1) =$	0.0890625	$P(Y=0 \mid Z_1=1, Z_2=1) =$
Normalized:	0.50831223	Normalize

Question A3 - Prediction

1. Does the model predict churn or no churn?
2. What is the predicted probability?

Model comparison and interpretation

1. "Why are the predicted churn probabilities so different? Because they used very different feature sets which show lesser differentiation on churn when or when not present. This resulted in model A having a much greater difference between the probabilities between churn and no churn."

$X_2 = 1$	$X_2 = 0$	Total
25	5	30
13	57	70

alpha = 1

$X_1 = 1$	$X_1 = 0$	Total
0.8125	0.1875	1
0.19444444	0.80555556	1

0.8125
0.19444444

0.02835648
0.1296082

The model does predict churn.

0.8703918

In particular, it looks like feature "Customer opened many support tickets" appeared to have a higher contribution to the resultant prediction of churn.

$Z_2 = 1$	$Z_2 = 0$	Total
15	15	30
28	42	70

alpha = 1

$Z_1 = 1$	$Z_1 = 0$	Total
0.5	0.5	1
0.40277778	0.59722222	1

0.5
0.40277778

0.08614969
0.49168777

The model very slightly predicts churn.
I would argue not enough to say with confidence.

The predicted difference

0.50831223

2. "Which model's prediction is more trustworthy? Why?"
Model A is more trustworthy due to its much greater difference between churn and not churn probabilities across the feature presence.
Compare, a predicted probability of ~0.87, versus ~0.49.
The latter is essentially a coinflip.

Problem 2:

Complete the Hidden Markov Model (HMM) Forward/Backtracking Table (reference in published slides on HMM) as we have gone through in the class. Using these transition and emission probabilities.

Raw Data:

Day	Weather (Hidden State)
1	Sunny
2	Sunny
3	Cloudy
4	Rainy
5	Sunny
6	Cloudy
7	Rainy
8	Rainy
9	Cloudy
10	Sunny

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Transition:

From -> To	Sunny	Cloudy	Rainy
Sunny	0.33	0.67	0
Cloudy	0.33	0	0.67
Rainy	0.33	0.33	0.33

Emission:

Weather-> Behavior	Walk	Umbrella
Sunny	1	0
Cloudy	0.66667	0.333333
Rainy	0.33333	0.666667

Forward Recurs.

Day	Observation
1	Walk
2	
	Walk
3	
	Walk
4	
	Umbrella
5	
	Walk
6	
	Umbrella
7	
	Walk

Friend's Behavior (Observation)

Walk	
Walk	
Walk	
Umbrella	
Walk	
Umbrella	
Walk	
Umbrella	
Walk	
Walk	

8	
	Umbrella
	V
9	
	Walk
	V
10	
	Walk
	V1

? -->	Sunny	Cloudy	Rainy	Day
V0(?)	0.333333333333333	0.333333333333333	0.333333333333333	
P(W ?)	1	0.666666666666667	0.333333333333333	1
V1(?)=V0(?)*P(W ?)	0.333333333333333	0.222222222222222	0.111111111111111	
V1(S)*P(? S)	0.11	0.223333333333333	0	2
V1(C)*P(? C)	0.073333333333333	0	0.148888888888889	
V1(R)*P(? R)	0.036666666666667	0.036666666666667	0.036666666666667	
P(U ?)	1	0.666666666666667	0.333333333333333	
V2(?)=max(?)*P(U ?)	Err:511	Err:511	Err:511	
V2(S)*P(? S)	Err:511	Err:511	Err:511	3
V2(C)*P(? C)	Err:511	Err:511	Err:511	
V2(R)*P(? R)	Err:511	Err:511	Err:511	
P(W ?)	1	0.666666666666667	0.333333333333333	
V3(?)=max(?)*P(W ?)	Err:511	Err:511	Err:511	
V3(S)*P(? S)	Err:511	Err:511	Err:511	4
V3(C)*P(? C)	Err:511	Err:511	Err:511	
V3(R)*P(? R)	Err:511	Err:511	Err:511	
P(W ?)	0	0.333333333333333	0.666666666666667	
V4(?)=max(?)*P(W ?)	Err:511	Err:511	Err:511	
V4(S)*P(? S)	Err:511	Err:511	Err:511	5
V4(C)*P(? C)	Err:511	Err:511	Err:511	
V4(R)*P(? R)	Err:511	Err:511	Err:511	
P(W ?)	1	0.666666666666667	0.333333333333333	
V5(?)=max(?)*P(W ?)	Err:511	Err:511	Err:511	
V5(S)*P(? S)	Err:511	Err:511	Err:511	6
V5(C)*P(? C)	Err:511	Err:511	Err:511	
V5(R)*P(? R)	Err:511	Err:511	Err:511	
P(W ?)	0	0.333333333333333	0.666666666666667	
V6(?)=max(?)*P(W ?)	Err:511	Err:511	Err:511	
V6(S)*P(? S)	Err:511	Err:511	Err:511	7
V6(C)*P(? C)	Err:511	Err:511	Err:511	
V6(R)*P(? R)	Err:511	Err:511	Err:511	
P(W ?)	1	0.666666666666667	0.333333333333333	
V7(?)=max(?)*P(W ?)	Err:511	Err:511	Err:511	

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V7(S)*P(? S)	Err:511	Err:511	Err:511	8
V7(C)*P(? C)	Err:511	Err:511	Err:511	
V7(R)*P(? R)	Err:511	Err:511	Err:511	
P(W ?)	0	0.333333333333333	0.666666666666667	
8(?)=max(?)*P(W ?)	Err:511	Err:511	Err:511	
V8(S)*P(? S)	Err:511	Err:511	Err:511	9
V8(C)*P(? C)	Err:511	Err:511	Err:511	
V8(R)*P(? R)	Err:511	Err:511	Err:511	
P(W ?)	1	0.666666666666667	0.333333333333333	
9(?)=max(?)*P(W ?)	Err:511	Err:511	Err:511	
V9(S)*P(? S)	Err:511	Err:511	Err:511	10
V9(C)*P(? C)	Err:511	Err:511	Err:511	
V9(R)*P(? R)	Err:511	Err:511	Err:511	
P(W ?)	1	0.666666666666667	0.333333333333333	
0(?)=max(?)*P(W ?)	Err:511	Err:511	Err:511	

Backtracking Reasoning	Ground Truth	Correct?
Day 1 was most likely Sunny. It was Sunny.	Sunny	Yes
The most likely Day 1 to cause this was... With Day 2 most likely Sunny. It was Sunny.	Sunny	Yes
The most likely Day 2 to cause this was... With Day 3 Most likely Cloudy. It was Cloudy.	Cloudy	Yes
The most likely Day 3 to cause this was... With Day 4 most likely Rainy. It was Rainy.	Rainy	Yes
The most likely Day 4 to cause this was... With Day 5 most likely Cloudy. It was Cloudy.	Sunny	No
The most likely Day 5 to cause this was... With Day 6 most likely Rainy. It was Rainy.	Cloudy	No
The most likely Day 6 to cause this was... With Day 7 most likely Cloudy.	Rainy	No

It was Cloudy.

The most likely Day 7 to cause this was...

With Day 8 Most likely Rainy	Rainy	Yes
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It was Rainy.

The most likely Day 8 to cause this was...

With Day 9 most likely Sunny	Cloudy	No
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It was sunny.

The most likely Day 9 to cause this was...

Day Ten was most likely Cloudy	Sunny	No
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